

Case Study 14 : SafeGuard Insurance - Policy & Claims Analytics

Industry: Insurance

Modules: Reporting, Exploration, Dashboard, Story Making

Real-Time Business Scenario : SafeGuard Insurance Company offers health, life, motor, and property insurance products. The company's Claims Processing team has flagged that claim settlement is taking longer than the IRDAI mandated 30-day period, rejection rates are higher than industry average, and fraudulent claims are rising. The Chief Actuary wants IBM Cognos Analytics to provide visibility into claims patterns, agent productivity, policy renewals, and fraud indicators to streamline operations.

Question 1: Reporting

Total Claims, Settled Claims, Pending Claims, Average Days to Settle, and Rejection Rate %. Apply Conditional Styling: Avg Days > 30 in Red (IRDAI breach), 20–30 in Amber, < 20 in Green. Add prompts for Agent Branch Create a Claims Settlement Report grouped by Policy Type showing and Policy Type. Header must include SafeGuard branding; Footer must include 'Regulatory Document'.

IBM Cognos Analytics InsuranceReport1

Report > Pages > Page1

Insertable objects

Find

Project14-dm

- CS14_Insurance_SafeGuard.xlsx
 - Policy ID
 - Policy Type
 - Premium Amount
 - Policy Start Date
 - Renewal Flag
 - Customer ID
 - Age Group
 - Gender
 - City
 - Agent Name
 - Agent Branch
 - Claim ID
 - Claim Date
 - Claim Amount
 - Claim Type
 - Settlement Date
 - Days to Settle
 - Claim Status
 - Fraud Indicator Flag
 - Rejection Reason
 - Customer Satisfaction Score
 - Agent Performance Rating

SafeGuard Insurance - Claims Settlement Report

Policy Type	Claim Status	Claim ID	Rejection Rate%	Days to Settle
Health	Pending	CLM00001	81.25	32
	Pending	CLM00004	81.25	25
	Pending	CLM00017	81.25	40
	Pending	CLM00018	81.25	25
	Pending	CLM00033	81.25	40
	Pending	CLM00048	81.25	25
	Pending	CLM00056	81.25	35
	Pending	CLM00058	81.25	40
	Pending	CLM00075	81.25	25
	Pending	CLM00091	81.25	30
	Pending	CLM00098	81.25	25
	Pending	CLM00117	81.25	25
	Pending	CLM00128	81.25	30
	Pending	CLM00131	81.25	30
	Pending	CLM00142	81.25	30
	Pending	CLM00164	81.25	28
	Pending	CLM00179	81.25	25
	Pending	CLM00181	81.25	30
	Pending	CLM00184	81.25	30
	Pending	CLM00188	81.25	35

Regulatory Document

Report > Pages > Page1				
Insertable objects				
Find				
Project14-dm				
CS14_Insurance_SafeGuard.xlsx				
Policy ID				
Policy Type				
Premium Amount				
Policy Start Date				
Renewal Flag				
Customer ID				
Age Group				
Gender				
City				
Agent Name				
Agent Branch				
Claim ID				
Claim Date				
Claim Amount				
Claim Type				
Settlement Date				
Days to Settle				
Claim Status				
Fraud Indicator Flag				
Rejection Reason				
Customer Satisfaction Score				
Agent Performance Rating				

Rejected	CLM00002	38.7755102	27
Rejected	CLM00003	38.7755102	58
Rejected	CLM00032	38.7755102	33
Rejected	CLM00045	38.7755102	56
Rejected	CLM00054	38.7755102	44
Rejected	CLM00064	38.7755102	60
Rejected	CLM00073	38.7755102	54
Rejected	CLM00076	38.7755102	39
Rejected	CLM00092	38.7755102	28
Rejected	CLM00095	38.7755102	60
Rejected	CLM00108	38.7755102	26
Rejected	CLM00120	38.7755102	52
Rejected	CLM00123	38.7755102	30
Rejected	CLM00130	38.7755102	29
Rejected	CLM00143	38.7755102	22
Rejected	CLM00170	38.7755102	3
Rejected	CLM00175	38.7755102	22
Rejected	CLM00186	38.7755102	63
Rejected	CLM00200	38.7755102	64
Property - Count		49	
Property - Average			34.26530612
Overall - Count		200	
Overall - Average			30.12

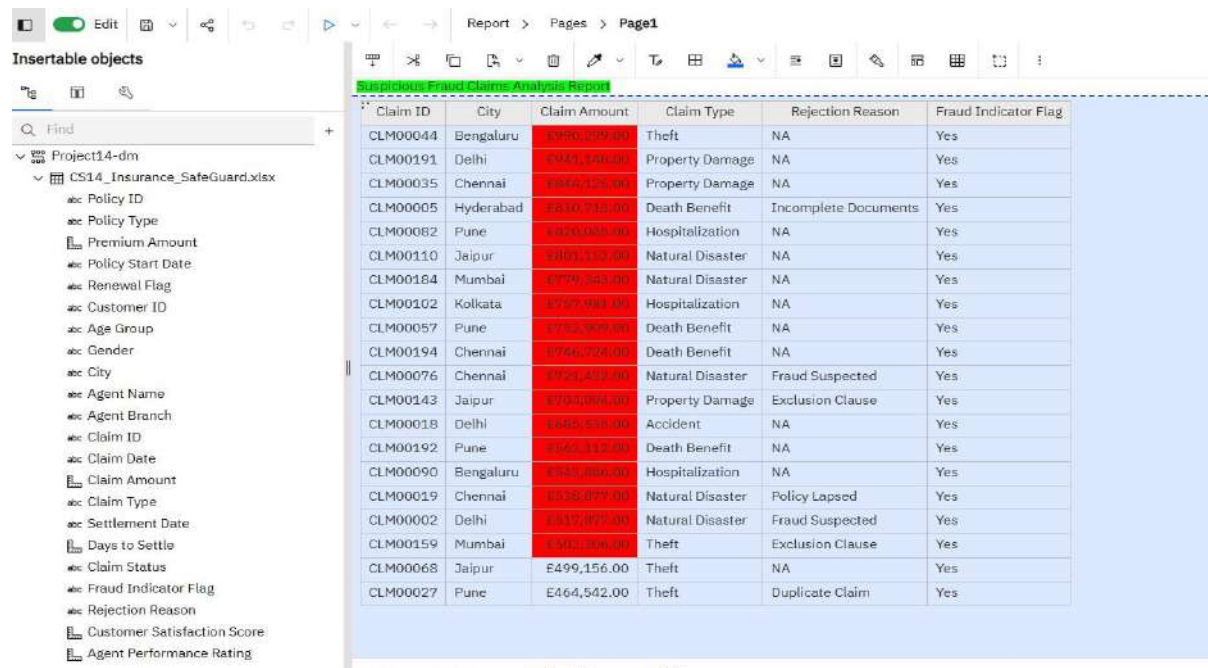
SafeGuard Insurance - Claims Settlement Report

Policy Type	Claim Status	Claim ID	Rejection Rate%	Days to Settle
<Policy Type>	<Claim Status>	<Claim ID>	<Rejection Rate%>	<Days to Settle>
<Policy Type> - Count		<Count(Claim ID)>		
<Policy Type> - Average				<Average(Days to Settle)>
<Policy Type>	<Claim Status>	<Claim ID>	<Rejection Rate%>	<Days to Settle>
<Policy Type> - Count		<Count(Claim ID)>		
<Policy Type> - Average				<Average(Days to Settle)>
Overall - Count		<Count(Claim ID)>		
Overall - Average				<Average(Days to Settle)>

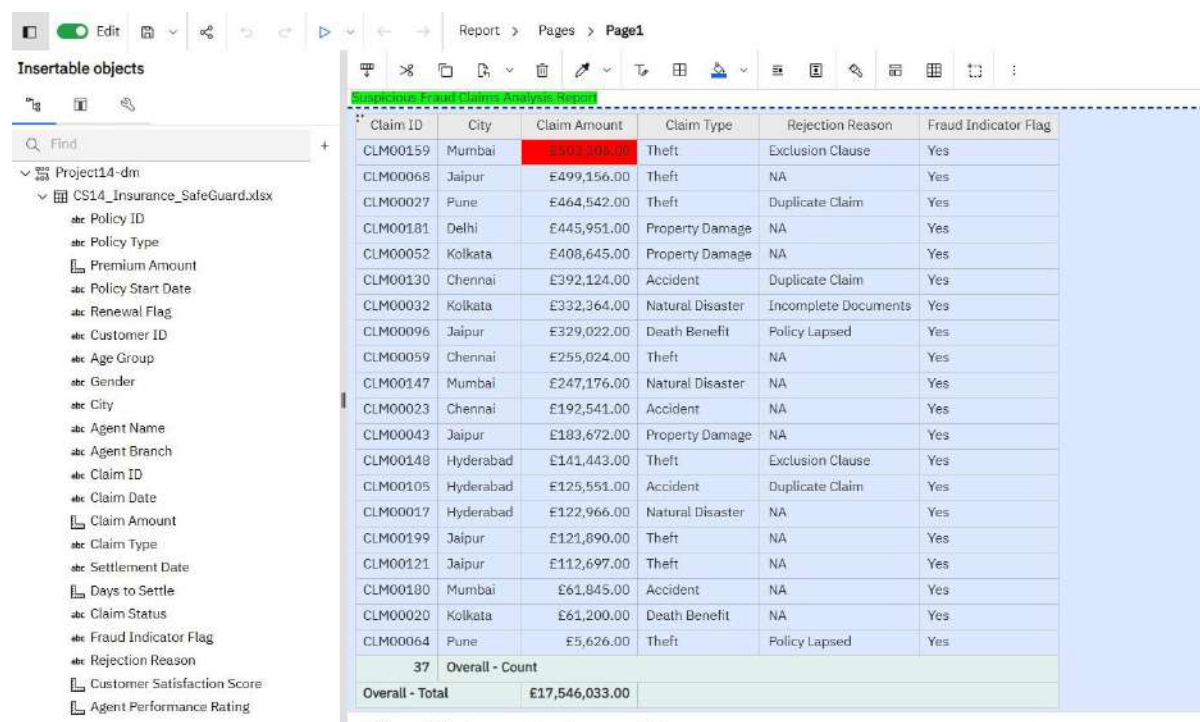
Regulatory Document

Question 2: Reporting

[Reporting] Build a Fraud Indicator Report listing all claims where Fraud Indicator = Yes, showing Claim ID, Customer City, Claim Amount, Claim Type, and Rejection Reason. Apply Conditional Styling: Claim Amounts > ₹5,00,000 flagged in Red Bold. Add a summary count and total suspicious claim value at the footer. Sort by Claim Amount descending.



Claim ID	City	Claim Amount	Claim Type	Rejection Reason	Fraud Indicator Flag
CLM00044	Bengaluru	₹990,000.00	Theft	NA	Yes
CLM00191	Delhi	₹947,540.00	Property Damage	NA	Yes
CLM00035	Chennai	₹864,325.00	Property Damage	NA	Yes
CLM00005	Hyderabad	₹830,715.00	Death Benefit	Incomplete Documents	Yes
CLM00082	Pune	₹820,000.00	Hospitalization	NA	Yes
CLM00110	Jaipur	₹801,370.00	Natural Disaster	NA	Yes
CLM00184	Mumbai	₹779,340.00	Natural Disaster	NA	Yes
CLM00102	Kolkata	₹757,980.00	Hospitalization	NA	Yes
CLM00057	Pune	₹752,500.00	Death Benefit	NA	Yes
CLM00194	Chennai	₹746,720.00	Death Benefit	NA	Yes
CLM00076	Chennai	₹721,470.00	Natural Disaster	Fraud Suspected	Yes
CLM00143	Jaipur	₹704,050.00	Property Damage	Exclusion Clause	Yes
CLM00018	Delhi	₹680,500.00	Accident	NA	Yes
CLM00192	Pune	₹667,310.00	Death Benefit	NA	Yes
CLM00090	Bengaluru	₹651,000.00	Hospitalization	NA	Yes
CLM00019	Chennai	₹638,870.00	Natural Disaster	Policy Lapsed	Yes
CLM00002	Delhi	₹632,870.00	Natural Disaster	Fraud Suspected	Yes
CLM00159	Mumbai	₹611,300.00	Theft	Exclusion Clause	Yes
CLM00068	Jaipur	₹499,156.00	Theft	NA	Yes
CLM00027	Pune	₹464,542.00	Theft	Duplicate Claim	Yes



Claim ID	City	Claim Amount	Claim Type	Rejection Reason	Fraud Indicator Flag
CLM00159	Mumbai	₹611,300.00	Theft	Exclusion Clause	Yes
CLM00068	Jaipur	₹499,156.00	Theft	NA	Yes
CLM00027	Pune	₹464,542.00	Theft	Duplicate Claim	Yes
CLM00181	Delhi	₹445,951.00	Property Damage	NA	Yes
CLM00052	Kolkata	₹408,645.00	Property Damage	NA	Yes
CLM00130	Chennai	₹392,124.00	Accident	Duplicate Claim	Yes
CLM00032	Kolkata	₹332,364.00	Natural Disaster	Incomplete Documents	Yes
CLM00096	Jaipur	₹329,022.00	Death Benefit	Policy Lapsed	Yes
CLM00059	Chennai	₹255,024.00	Theft	NA	Yes
CLM00147	Mumbai	₹247,176.00	Natural Disaster	NA	Yes
CLM00023	Chennai	₹192,541.00	Accident	NA	Yes
CLM00043	Jaipur	₹183,672.00	Property Damage	NA	Yes
CLM00148	Hyderabad	₹141,443.00	Theft	Exclusion Clause	Yes
CLM00105	Hyderabad	₹125,551.00	Accident	Duplicate Claim	Yes
CLM00017	Hyderabad	₹122,966.00	Natural Disaster	NA	Yes
CLM00199	Jaipur	₹121,890.00	Theft	NA	Yes
CLM00121	Jaipur	₹112,697.00	Theft	NA	Yes
CLM00180	Mumbai	₹61,845.00	Accident	NA	Yes
CLM00020	Kolkata	₹61,200.00	Death Benefit	NA	Yes
CLM00064	Pune	₹5,626.00	Theft	Policy Lapsed	Yes
37 Overall - Count					
Overall - Total		₹17,546,033.00			

Suspicious Fraud Claims Analysis Report

Claim ID	City	Claim Amount	Claim Type	Rejection Reason	Fraud Indicator Flag
<Claim ID>	<City>	<Claim Amount>	<Claim Type>	<Rejection Reason>	<Fraud Indicator Flag>
<Claim ID>	<City>	<Claim Amount>	<Claim Type>	<Rejection Reason>	<Fraud Indicator Flag>
<Claim ID>	<City>	<Claim Amount>	<Claim Type>	<Rejection Reason>	<Fraud Indicator Flag>
<Count(Claim ID)>	Overall - Count				
Overall - Total		<Total(Claim Amount)>			

Question 3: Reporting

[Reporting] Using Report Visualize Prompt, allow Regional Managers to select chart type (Bar or Column) to view Monthly Claims Volume vs Settlement Rate. Add prompts for Policy Type and Date Range. Summarize months where settlement rate dropped below 70% using Conditional Styling in Amber. Add a text annotation of the top 3 settlement bottlenecks in the report.

Report > Pages > Page1

Insertable objects

Find

Project14-dm

CS14_Insurance_SafeGuard.xlsx

Policy ID

Policy Type

Premium Amount

Policy Start Date

Renewal Flag

Customer ID

Age Group

Gender

City

Agent Name

Agent Branch

Claim ID

Claim Date

Claim Amount

Claim Type

Settlement Date

Days to Settle

Claim Status

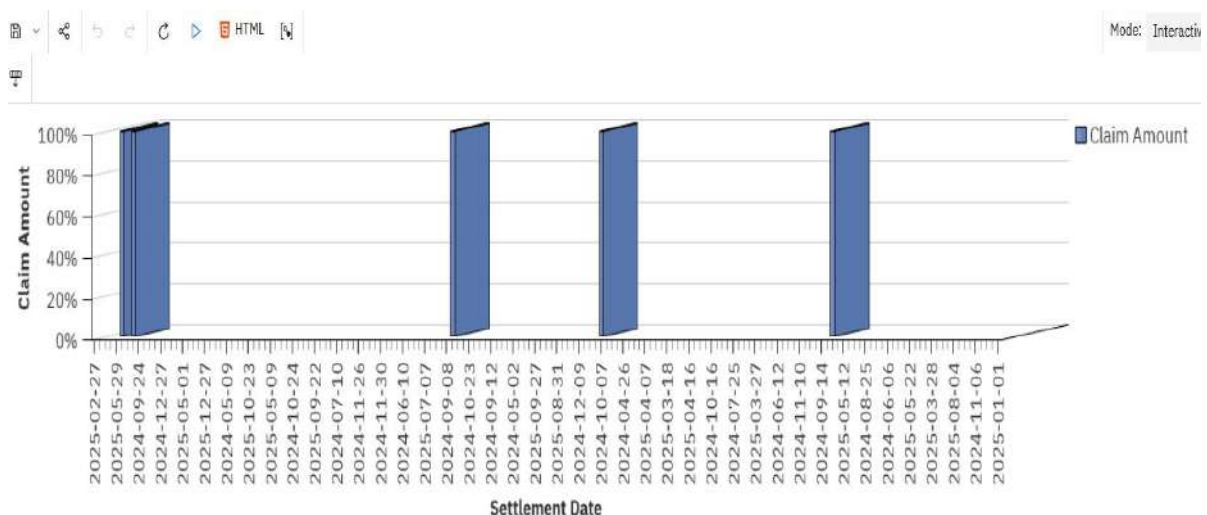
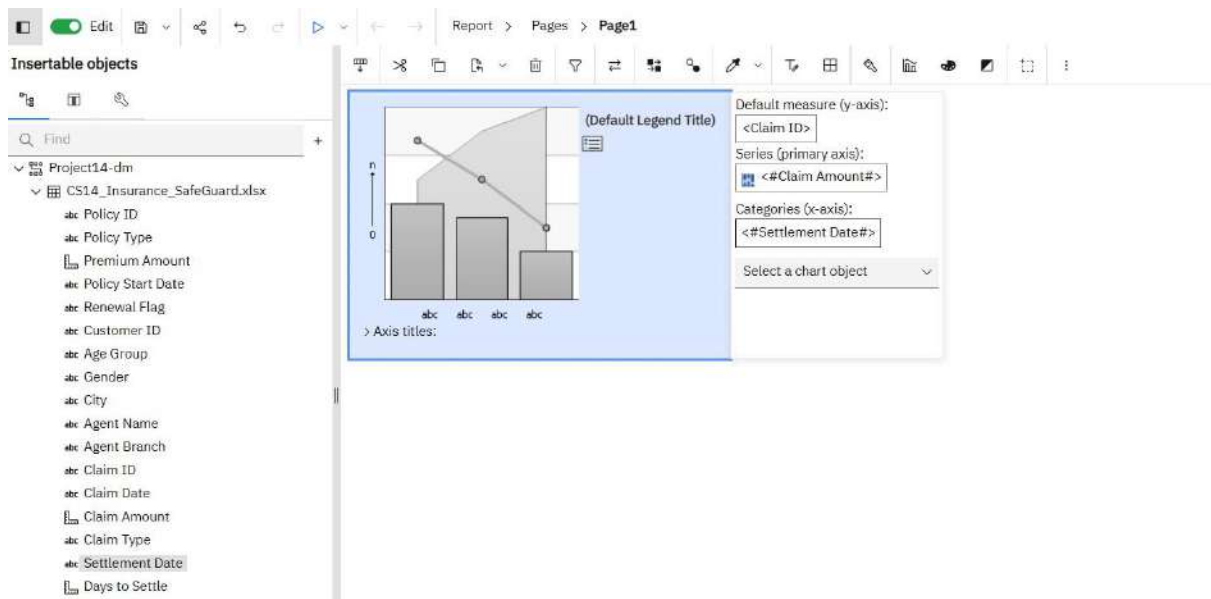
Fraud Indicator Flag

Rejection Reason

Customer Satisfaction Score

Agent Performance Rating

Claim ID	Claim Date	Settlement Date	Claim Status
CLM00181	2025-03-12	2025-04-19	Pending
CLM00182	2024-08-19	2024-08-20	Rejected
CLM00183	2024-12-08	2024-12-15	Settled
CLM00184	2025-02-23	2025-04-23	Pending
CLM00185	2024-11-28	2024-12-09	Rejected
CLM00186	2024-05-19	2024-07-10	Rejected
CLM00187	2025-04-30	2025-05-19	Pending
CLM00188	2025-08-24	2025-10-08	Pending
CLM00189	2024-05-22	2024-05-28	Pending
CLM00190	2025-08-24	2025-08-29	Rejected
CLM00191	2024-06-11	2024-06-12	Settled
CLM00192	2025-08-28	2025-09-11	Settled
CLM00193	2024-09-09	2024-09-11	Settled
CLM00194	2024-11-19	2025-01-01	Settled
CLM00195	2024-10-05	2024-12-02	Pending
CLM00196	2025-06-13	2025-07-28	Settled
CLM00197	2025-01-05	2025-02-23	Pending
CLM00198	2025-04-23	2025-06-04	Pending
CLM00199	2025-01-21	2025-01-25	Settled
CLM00200	2025-04-24	2025-06-17	Rejected
Overall - Calculated		2024-03-24	
Overall - Calculated		2024-03-24	



Question 4: Exploration

Use Cognos Exploration to analyze SafeGuard's claims dataset. Explore whether Age Group or City correlates with higher claim frequency and fraud flags. Investigate the relationship between Agent Performance Rating and Claim Rejection Rate. Identify which Policy Types have the longest average settlement times and what drives the delays.

1. Age Group vs Claims & Fraud